

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	B2
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: AI-Driven Enterprise Knowledge Digital Twin using Agentic AI and Hybrid RAG

Modern organizations generate a large volume of enterprise documents such as HR policies, employee handbooks, standard operating procedures (SOPs), project documentation, technical manuals, and compliance guidelines. Employees often spend significant time searching for the required information, and existing enterprise search systems mainly rely on keyword-based retrieval or basic AI-assisted search, which may not always provide accurate, context-aware, or explainable responses. This leads to reduced productivity and inefficient knowledge management.

This project proposes an **AI-Driven Enterprise Knowledge Digital Twin using Agentic AI and Hybrid Retrieval-Augmented Generation (Hybrid RAG)** to provide an intelligent enterprise knowledge management platform. The system creates a digital representation of an organization's knowledge by understanding enterprise documents and enabling users to retrieve information using natural language queries. It employs multiple AI agents to perform authentication, query planning, intelligent document retrieval, reasoning, response verification, and citation generation. Hybrid RAG combines keyword-based and semantic search techniques to improve retrieval accuracy and reduce irrelevant results.

The proposed system also includes role-based access control to ensure secure access to organizational documents. Every response is supported with source citations, page numbers, and confidence scores to improve transparency and reduce AI hallucinations. Additional features such as document version management, knowledge analytics, duplicate document detection, and knowledge gap identification further enhance enterprise knowledge management.

The scope of this project is limited to intelligent enterprise document retrieval and question answering using organizational knowledge. It does not aim to replace enterprise resource planning systems or automate organizational decision-making. The expected outcome is a secure, scalable, and explainable AI-powered knowledge platform that reduces document search time, improves retrieval accuracy, enhances employee productivity, and provides reliable responses with proper evidence.

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References (Category specific)

Reference 1: AgenticRAG: Agentic Retrieval for Enterprise Knowledge Bases

Authors: Microsoft Research

Year: 2026

Link:

<https://arxiv.org/abs/2605.05538>

This paper proposes an Agentic Retrieval-Augmented Generation framework for enterprise knowledge bases. Unlike traditional RAG, it uses multiple AI agents to perform planning, retrieval, reasoning, and verification, significantly improving answer quality and retrieval accuracy. This paper serves as the primary inspiration for our project's multi-agent enterprise knowledge architecture.

Reference 2: Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) for Enterprise Knowledge Management and Document Automation: A Systematic Literature Review

Journal: Applied Sciences (MDPI)

Year: 2026

Link:

<https://www.mdpi.com/2076-3417/16/1/368>

This systematic literature review analyzes the application of Retrieval-Augmented Generation and Large Language Models in enterprise knowledge management. It identifies current challenges, research gaps, and future directions, providing a strong foundation for developing intelligent enterprise knowledge systems.

Reference 3: Retrieval-Augmented Generation for Natural Language Processing

Publisher: Springer

Year: 2026

Link:

<https://link.springer.com/article/10.1007/s10462-026-11605-7>

This survey presents modern Retrieval-Augmented Generation techniques, vector databases, retrieval strategies, evaluation metrics, and challenges. It provides comprehensive insights into Hybrid RAG systems used in intelligent information retrieval

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